



TerraSatch AvyTS Module

Avalanche Terrain Intelligence + SherpAI Operational Agent

Version v1.1.0 - March 05, 2026

Prepared for avalanche forecasters (professional + volunteer), ski patrol, guiding operations, transportation corridor managers, SAR, land managers, backcountry educators, and technical partners.

Website: <https://terrasatch.com>

Data Portal: <https://data.terrasatch.com>

0. Front Matter

0.1 Title Page (version, date)

Document Title: TerraSatch AvyTS Module: Avalanche Terrain Intelligence + SherpAI Operational Agent

Version: v1.1.0

Date: March 05, 2026

0.2 Disclaimer (safety/legal)

TerraSatch provides decision-support software intended to improve terrain awareness and operational coordination. AvyTS is a decision-support module focused on avalanche and snowpack operations. TerraSatch does not prevent accidents and does not replace professional training, certified avalanche education, local advisories, or organizational safety protocols.

Avalanche terrain, weather, and snowpack involve uncertainty, incomplete information, and rapidly changing conditions. Users must verify outputs against field observations, official forecasts and warnings, and site-specific expertise. AvyTS and SherpAI may summarize and highlight patterns, but they do not determine “safe” routes or certify stability.

This document describes AvyTS V1 scope, operating model, and implementation approach. Any future items are described as planned directions rather than commitments.

0.3 Links

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1. Executive Summary

AvyTS is TerraSatch's avalanche and snowpack module: a terrain-intelligence workflow that turns fragmented evidence (weather inputs, field observations, pits and tests, avalanche events, and remote sensing products) into a current, auditable operational picture. AvyTS is built for real-world cadence: early-morning synthesis, fast field capture, mid-day updates, and after-action review. SherpAI is the operational agent that accelerates ingestion, QA/QC, summarization, and product drafting while enforcing safety language guardrails and approval gates.

V1 is intentionally conservative. It prioritizes evidence linkage, explicit uncertainty, and repeatability over automation. Any external distribution remains human-approved.

1.1 V1 operating principles

Clarity	Use consistent terminology, time windows, and zone definitions. Separate observations from inferences. Provide "what changed" and "why it matters."
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Uncertainty	Express confidence with reasons. Identify data gaps, representativeness limits, and competing hypotheses. Trigger review friction when evidence is sparse/conflicting.
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Traceability	Every conclusion points back to record IDs, timestamps, and sources. Every publication creates a snapshot that can be reconstructed for retrospectives.
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1.2 Scope boundaries

In scope (V1)	Offline-first field capture; standardized snowpit and avalanche observation schemas; media attachments; ingestion + normalization + QA/QC; terrain derivatives (slope/aspect/elevation); map-first triage, time windowing, reporting, and exports; ingestion of common drone deliverables (orthomosaic GeoTIFFs, DEM/DSM, point clouds where provided); connectivity-aware sync (cellular + satellite backhaul); alerts engine (push + in-app); SherpAI draft briefs and anomaly flags; role-gated publishing and audit logs.
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Out of scope (V1)	Automated go/no-go decisions; route endorsements; guarantees of safety; proprietary access claims to external data; black-box inference without provenance; universal calibrated models across all regions without local validation.
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2. The Avalanche Operations Problem

2.1 Fragmented information and brittle workflows

Operational inputs are split across websites, PDFs, spreadsheets, radios, and chat threads. This fragmentation drives missed signals, duplicated effort, and inconsistent language—especially during high-tempo storm cycles.

2.2 Terrain-scale complexity and microclimates

Slope-scale variation matters. Aspect, elevation, wind exposure, canopy, and terrain traps change the outcome. A system must connect evidence to terrain context and preserve representativeness limits.

2.3 Communication risk under time pressure

Operations require repeated briefing, re-briefing, and partner communication. Misstated confidence or ambiguous wording can propagate quickly. V1 enforces safe language and approval gates to reduce miscommunication risk.

3. AvyTS Overview

3.1 What AvyTS is

AvyTS is an evidence-driven avalanche operations system: it standardizes capture (observations, pits/tests, incidents), attaches terrain context, runs QA/QC, supports map-based triage and filtering, drafts products, and preserves auditability.

3.2 What AvyTS is not

AvyTS is not an official bulletin, not a guarantee, and not a route-approval engine. It does not replace local advisories, patrol/guide SOPs, or professional judgment.

3.3 TerraGrid and the terrain foundation

TerraSatch's terrain model (TerraGrid) provides the basemap, slope/aspect/elevation derivatives, and consistent spatial indexing. AvyTS attaches this context to every record and output so decisions remain tied to terrain reality.

4. SherpAI + AvyTS: How They Work Together

4.1 Role and responsibilities

SherpAI is an operational co-forecaster assistant. It accelerates ingestion and synthesis but does not replace human decision-making. SherpAI's outputs are drafts and flags until a human approves publication.

4.2 SherpAI use cases in AvyTS

- 1) Morning synthesis: pull the last 24–48 hours of evidence, summarize “what changed,” propose hypotheses, and list verification priorities.
- 2) Field triage: as observations arrive, group duplicates, flag contradictions, and route urgent items to reviewers.
- 3) Snowpit comparison: identify similar layer structures across locations, highlight recurring weak layers, and draft stratigraphy summaries.
- 4) Incident packaging: generate an incident packet with map snapshot, timeline, measurements, and evidence links; enforce sensitivity rules.
- 5) Partner corridor brief: draft DOT-style briefs focused on segments, triggers, and uncertainty; never claim closures or mandates.
- 6) Retrospectives: compile “cycle story” reports linking hypotheses to later observations and events.

4.3 Guardrails and approval gates

Allowed language	“Based on reported observations up to [time], the primary concern is [problem]. Confidence is [low/medium/high] due to [gaps].” “Decision support; reference official bulletins and verify in the field.”
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Forbidden language	“This route is safe.” “No avalanche risk.” “Stability is confirmed.” Any route endorsement language.
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All externally shared products require an authorized human approval action that is recorded in the audit log.

5. Full Scope of AvyTS Operations

5.1 Daily cadence

0600–0800: ingest overnight station data and forecast products; review new observations; SherpAI drafts the morning brief; forecaster approves internal distribution.

0800–1200: field operations: capture obs and pits; update review queue; optional drone missions run where needed; urgent alerts triaged.

1200–1600: mid-day update: incorporate new evidence, update overlays, publish partner briefs as authorized; incident packet generation if required.

Post-ops: archive snapshot, note decisions, and stage retrospective questions (what evidence supported what call).

5.2 Roles and permissions

Org Admin configures regions, zones, sharing, and alert rules. Forecasters/Ops Leads approve publications and manage sensitivity. Field Observers capture and submit records. External Viewers receive approved read-only outputs.

6. Inputs and Data Sources (V1)

6.1 Weather inputs

Commonly used sources include automated weather stations, forecast grids, and model guidance. AvyTS stores raw values, derived features (e.g., storm totals, wind windows), and provenance metadata.

6.2 Snowpack and field observations

Field observations include travel/route notes, red flags, surface conditions, avalanche events, and structured pit/test records. AvyTS preserves both structured fields and narrative text, with optional controlled vocabulary mapping.

6.3 Terrain and basemaps

Terrain sources include DEMs and LiDAR-derived surfaces where available. AvyTS uses slope/aspect/elevation bands, terrain traps, and exposure proxies as context layers—not deterministic truth.

6.4 Remote sensing products

AvyTS can ingest drone deliverables (orthomosaics, DEM/DSM, point clouds) and other raster/vector layers produced by partner workflows. Inputs are treated as evidence layers with mission metadata and lineage.

7. Drone Operations and Remote Sensing

7.1 Operational goals for drones in AvyTS

Drones support: (a) safer evidence collection (crowns/debris), (b) updated terrain context after storms or slides, (c) repeat-mission change awareness, and (d) documentation for incident packets.

7.2 Deliverables and ingestion

V1 expects standard deliverables exported from established drone processing workflows (e.g., orthomosaic GeoTIFFs; DEM/DSM; optional LAS/LAZ). AvyTS stores:

mission time window, sensor type, resolution, processing software/version (if provided), georeferencing, and QA flags.

7.3 Drone-to-brief workflow (ASCII)

```
Mission plan -> Capture -> Export deliverables -> Ingest as evidence layer
Evidence layer -> QA/QC (projection, resolution, bounds) -> Attach to zone/time window
SherpAI -> Draft "what changed" notes -> Human review -> Publish snapshot + partner packet
```

8. Deployment and Off-Grid Connectivity

8.1 Offline-first capture

Field clients preload zones and tiles, capture records locally, and sync when connectivity is available.

8.2 Satellite backhaul and remote deployments

Satellite connectivity (e.g., LEO backhaul) is treated as a transport layer. V1 sync is bandwidth-aware: structured metadata syncs first; media uploads are resumable and can be deferred.

8.3 Edge posture

AvyTS supports hybrid operation: field capture at the edge and centralized processing in the cloud. Organizations can define data residency and sharing constraints per tenant policy.

9. Real-Time Notifications and Alerting

9.1 Channels

V1 supports push notifications (mobile), in-app alerts (web + mobile), and optional downstream channels (email/SMS/webhooks) where enabled by policy and agreements.

9.2 Alert rule types

Alerts can be triggered by: new avalanche events in watched zones; new high-salience red flags; rapid loading thresholds; mission ingestion completion; QA/QC anomalies; and review queue SLA breaches.

9.3 Safety, throttling, and auditability

Alerts avoid definitive safety language and include evidence links, time window, and a confidence/gap note. Alert rules, deliveries, and acknowledgements are logged for accountability and tuning.

10. Product Outputs (V1)

10.1 Operational dashboard

Map-first dashboard with filters, time windows, review queue, evidence attachments, and snapshot-based reporting.

10.2 Brief products (drafted by SherpAI; approved by humans)

Morning brief: what changed, key problems by elevation/aspect, confidence + gaps, verification plan.

Mid-day update: delta since morning, new evidence, changes in confidence, actions/notes per SOP.

Partner brief: corridor/segment-focused watch items, triggers, and uncertainty (no mandates).

10.3 Incident packets

Snapshot, timeline, measurements, media, and evidence linkage with redaction controls.

11. Risk Communication and Snow Science Visuals

11.1 Risk strip (orientation; not an official bulletin)

The risk strip is used to orient users to common terminology and to reinforce decision-support framing. It is not an official danger rating product.



11.2 Enhanced SnowPylot-style grid (illustrative)

AvyTS supports a grid-style view to summarize relative concern by elevation band × aspect bin for a time window. This is a decision-support visualization.

11.3 Stratigraphic column (illustrative)

Structured pit records can be rendered into a stratigraphic column for fast comparison across sites. Example format shown below.

12. Technical Architecture and API Surfaces (V1)

12.1 End-to-end flow

Field capture -> Sync Gateway -> Event Store
 Event Store -> QA/QC + Normalization -> Review Queue
 Review Queue -> SherpAI Drafts + Flags -> Human Approval
 Human Approval -> Publish Snapshots -> Map/Reports/Exports
 Events + Snapshots -> Alerts Engine -> Push/In-app notifications

12.2 Representative API surfaces

Category	Representative endpoints (illustrative)
Auth	POST /auth/login POST /auth/refresh GET /me
Events	POST /v1/events GET /v1/events?filters=... GET /v1/events/{id}
Observations	POST /v1/observations GET /v1/observations?time=...&zone=...
Incidents	POST /v1/incidents GET /v1/incidents/{id}
Layers/Tiles	GET /v1/layers GET /v1/tiles/{z}/{x}/{y}?layer=...
Reports	POST /v1/reports/daily-brief GET /v1/reports/{id}
Alerts	POST /v1/alerts/rules POST /v1/alerts/subscriptions POST /v1/alerts/test

Endpoints are shown for clarity; actual paths may vary by deployment.

13. Data Schemas

13.1 Observation intake JSON (example)

```
{
  "schema_version": "1.0",
  "type": "observation",
  "observed_at_utc": "2026-03-05T15:12:44Z",
  "observer": {
    "user_id": "usr_0042",
    "role": "field_observer"
  },
  "location": {
    "type": "Point",
    "coordinates": [
      -111.6543,
      40.5812
    ],
    "elevation_m": 3050
  },
  "terrain_context": {
    "aspect_bin": "NW",
    "slope_angle_deg": 36,
    "elevation_band": "upper"
  },
}
```

```

"category": "snowpit",
"tests": [
  {
    "type": "ECT",
    "result": "ECTP",
    "score": 12,
    "fracture_character": "Q2"
  }
],
"layers_summary": {
  "suspected_pwl": true,
  "notes": "Thin weak layer 60\u2013370 cm below surface in this pit."
},
"red_flags": [
  "shooting_cracks"
],
"summary": "Reactive wind-loaded pockets near ridgeline; cracking observed in adjacent terrain.",
"media": [
  {
    "type": "photo",
    "caption": "Crown line and deposition",
    "privacy": "internal"
  }
],
"qa_qc": {
  "flags": [
    "needs_review"
  ],
  "review_status": "pending"
}
}

```

13.2 Avalanche event / incident JSON (example)

```

{
  "schema_version": "1.0",
  "type": "avalanche_event",
  "occurred_at_utc": "2026-03-05T20:05:00Z",
  "reporter": {
    "user_id": "usr_0017",
    "role": "forecaster"
  },
  "location": {
    "type": "Point",
    "coordinates": [
      -111.6528,
      40.5797
    ],
    "elevation_m": 3120
  },
  "classification": {
    "trigger": "human",
    "problem_hypothesis": "wind_slab",
    "size": {
      "destructive_size": "D2",
      "relative_to_path": "R2"
    },
    "confidence": "medium"
  },
  "measurements": {
    "crown_height_cm": 60,

```

```

    "width_m": 40,
    "run_length_m": 250
  },
  "impacts": {
    "involvement": "none_reported",
    "notes": "No injuries reported."
  },
  "links": {
    "related_observations": [],
    "media": []
  },
  "qa_qc": {
    "flags": [
      "supervisor_review"
    ],
    "review_status": "pending"
  }
}

```

14. Use Cases (V1)

14.1 Ski patrol and controlled terrain operations

Shift briefs; controlled internal comms; documenting mitigation results; tracking slab type changes; incident packets for internal review; and limited partner distribution where authorized.

14.2 Guiding operations

Route planning support framed as hypotheses with confidence; client-safe summaries; tracking recurring weak layers; evidence packaging for guide teams across multiple zones.

14.3 Transportation corridors (DOT) and canyon operations

Segment-focused briefs; watch items and triggers tied to evidence; rapid updates during storm cycles; sensitivity controls for public release; archived snapshots for post-event review.

14.4 Search and rescue coordination

Rapid incident packaging with map snapshot + timeline; evidence and attachment bundling; controlled sharing; post-mission debrief reconstruction.

14.5 Education and public communication

Non-prescriptive summaries emphasizing uncertainty, official bulletin references, and evidence explanations. AvyTS can support educators by illustrating how evidence supports hypotheses without providing route endorsements.

14.6 Drone-supported incident documentation

When drone missions are used, AvyTS stores deliverables as evidence layers and links them to incidents and briefs to reduce confusion and improve traceability.

15. Roadmap

15.1 Phase 0 (current baseline)

Schemas, offline capture, ingestion + QA/QC, dashboard, report templates, alerts engine baseline, and audit logs.

15.2 Phase 1 (V1 operational release)

Hardened sync and review workflows, role-gated publishing, stable exports, and partner pilot onboarding playbooks.

15.3 Phase 2 (planned directions)

Repeat-mission change detection, stronger calibration and verification tooling, expanded integrations, and region onboarding automation.

16. References

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